

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)	(i)	Change in flux linkage or cutting flux (1) Complete circuit or implied (1)	2			2		
		(ii)	These are in the plane of motion/the vertical component is perpendicular to motion/these are not cut		1		1		
	(b)		Use of $\frac{BAN}{t}$ or $V = Blv$ (1) Use of $I = \frac{V}{R}$ (1) Answer = $62 \mu\text{T}$ (1)	1 1	 1		 3	3	
	(c)		Current down in resistor [or up in conductor or anticlockwise] (1) [F]RHR or [F]LHR (for electrons) or RH grip rule (1)	1	1		2		
	(d)		Other force compared e.g. friction/drag (1) Magnetic force small / resistive force is much bigger (1) Because current small / B small (1) Good calculation e.g. $P = I^2R = 8.8 \times 10^{-7} \text{ [W]}$ or $F = BIL = 1.6 \times 10^{-8} \text{ [N]}$ ecf on B or calculated values of B using I (1)			4	4	1	
			Question 5 total	5	3	4	12	4	0